

2026 | By: Shubham Tiwari



Internship Project Report

Power BI Social Media Analytics Dashboard

Cover Page

Objective

The objective of this project is to analyze social media engagement using Power BI dashboards. The dashboards provide insights into tweet performance, engagement trends, media interactions, and user activities to support data-driven decision-making.

Dataset

- Social Media Dataset (<https://data.world/markbradbourn/rwfd-real-world-fake-data/workspace/file?filename=SocialMedia.csv>)
- CSV Format
- Twitter Engagement Data
- Imported into Power Bi
- Problem statement by Elevance Skills

Tools Used

Tools

- Power bi
- Power Query
- Dax
- Csv data set

Purpose

Dashboard development
Data Cleaning
Visual Filter
Data Source

Data Preparation

Include points like:

- Imported CSV dataset
- Removed unnecessary columns
- Changed data types

- Removed null values
- Created calculated columns
- Created DAX measures
- Applied filters

Filters

Search

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

Visualizations

Build visual

Values

Add data fields here

Drill through

Cross-report

Off

Keep all filters

On

Add drill-through fields here

Data

Search

SocialMedia (1)

☐ Chart Visibility

☐ Σ Day number

☐ Even Day

☐ Σ hashtag clicks

☐ Σ Hour

☐ Σ Interaction

☐ Σ media engage...

☐ time

☐ Time Filter

☐ Tweet

☐ Tweet cateogry

☐ Σ url clicks

☐ Σ user profile clicks

☐ Σ Word Count

Dashboard 1

Tweet Interaction Breakdown by Category

Objective

Analyze URL clicks, Profile clicks, and Hashtag clicks based on tweet categories.

Problem statement

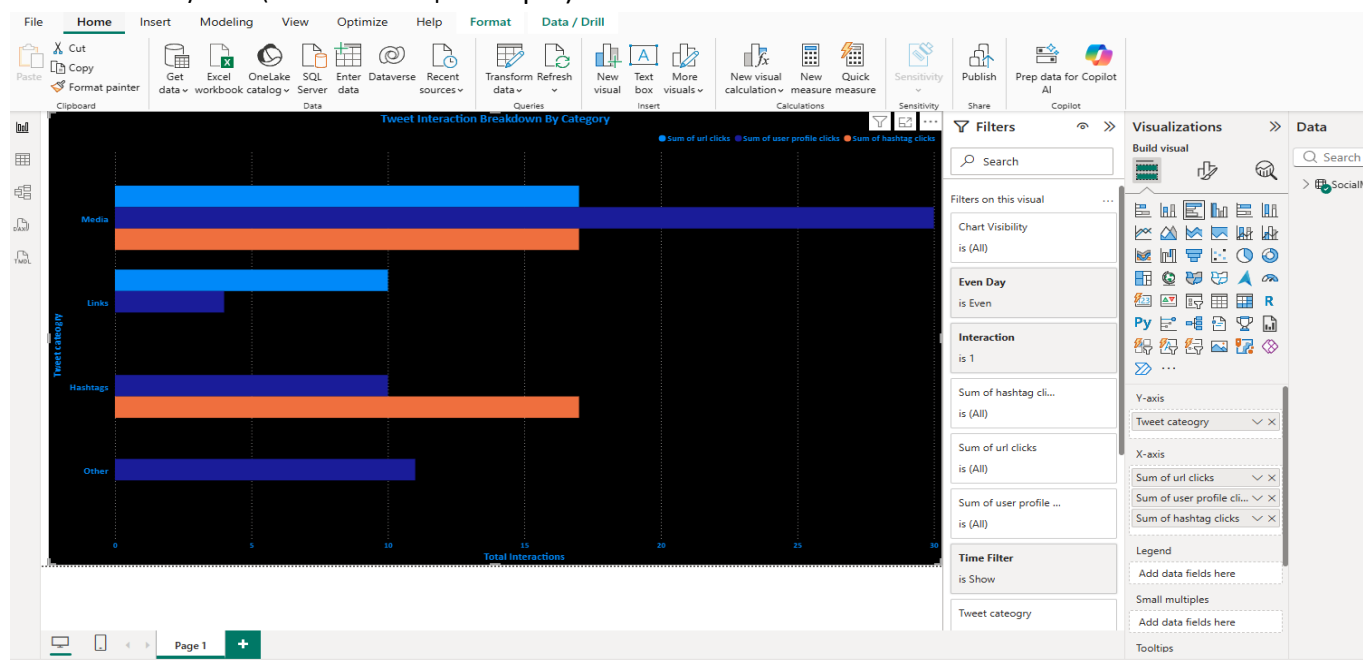
Tweet Interaction Breakdown by Category Create a clustered bar chart that shows the sum of URL clicks, user profile clicks, and hashtag clicks grouped by tweet category such as tweets with media, links, or hashtags. Include only tweets that have at least one of these interaction types. Apply additional conditions where the visualization should only be visible between 3 PM IST and 5 PM IST, and hidden at other times. Also, filter the dataset to include only tweets where the tweet date is an even number and the tweet word count is greater than 40. This task helps analyze how different types of tweets drive user interactions.

Visual Used

Clustered Bar Chart

Key Insights

- Media tweets generated higher engagement.
- URL clicks were lower than profile clicks.
- Hashtag interactions varied by category.
- Visibility filter (will visible on 3pm to 5pm)



Dashboard 2

Engagement Rate Comparison Analysis

Objective

To compare the average engagement rate across tweets and evaluate which content type receives better audience engagement.

Problem statement

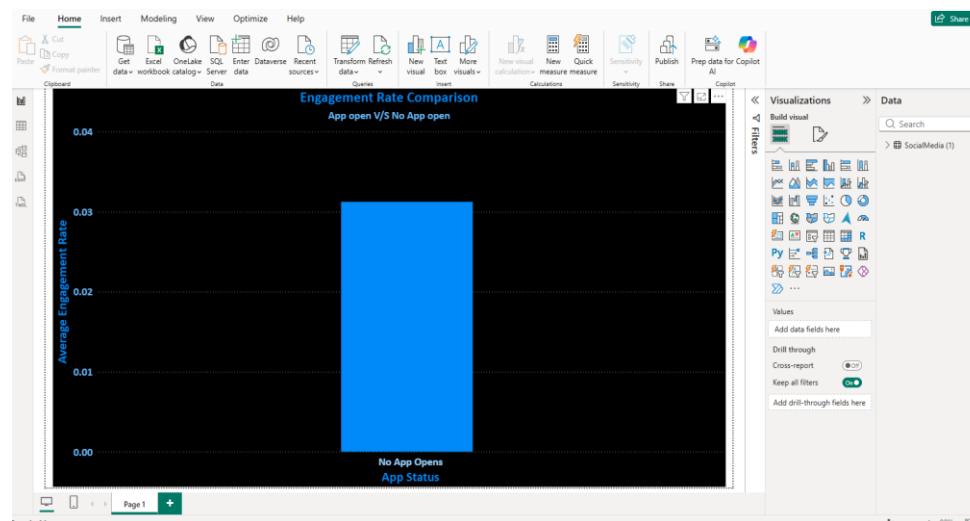
Engagement Rate Comparison Analyze and compare the engagement rate of tweets that include app opens versus those that do not. Include only tweets posted between 9 AM and 5 PM on weekdays. The visualization should be active only between 12 PM IST to 6 PM IST and 7 AM to 11 AM IST. Apply filters where tweet impressions are even numbers, tweet dates are odd numbers, and tweet character count is above 30. Also, remove tweets that contain the letter 'D'. This task helps evaluate how app interactions influence engagement rates.

Visualization Used

- Bar Chart

Key Insights

- content types.
- Engagement rates differed among tweet categories.
- Tweets with higher interaction generated better engagement rates.
- Average engagement provides a useful measure of tweet performance.
- The analysis highlights the most effective



Dashboard 3

Objective

To analyze media views and media engagements by day of the week and identify the days with the highest audience interaction.

Problem statement

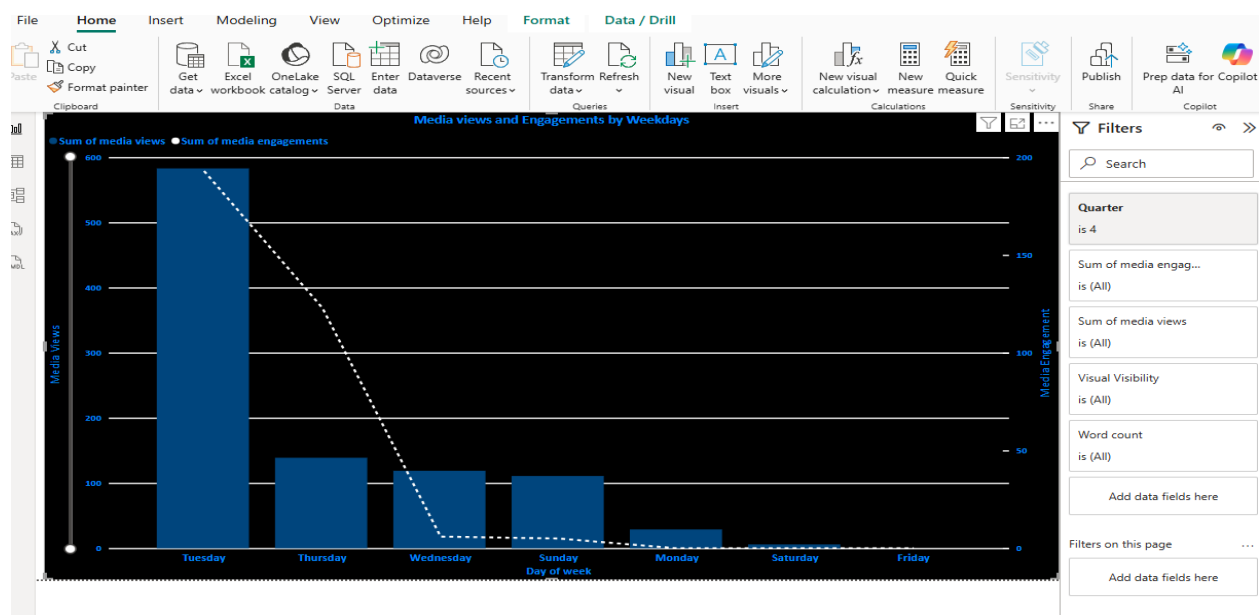
Media Interaction by Day of Week Create a dual-axis chart that visualizes both media views and media engagements by day of the week for the last quarter. Highlight days where there are noticeable spikes in interactions. The chart should only be visible between 3 PM IST to 5 PM IST and 7 AM to 11 AM IST. Apply filters such that tweet impressions must be even numbers, tweet dates must be odd numbers, and tweet character count must be above 30. Additionally, exclude tweets that contain the letter 'H' in their text. This task focuses on identifying patterns in media engagement across different days.

Visualization Used

- Line and Clustered Column Chart

Key Insights

- Media views and engagements changed across the week.
- Certain days recorded higher audience activity.
- Media engagement generally followed the trend of media views.



Dashboard 4

Objective

To compare the total number of replies, retweets, and likes received by tweets during the selected time period.

Problem statement

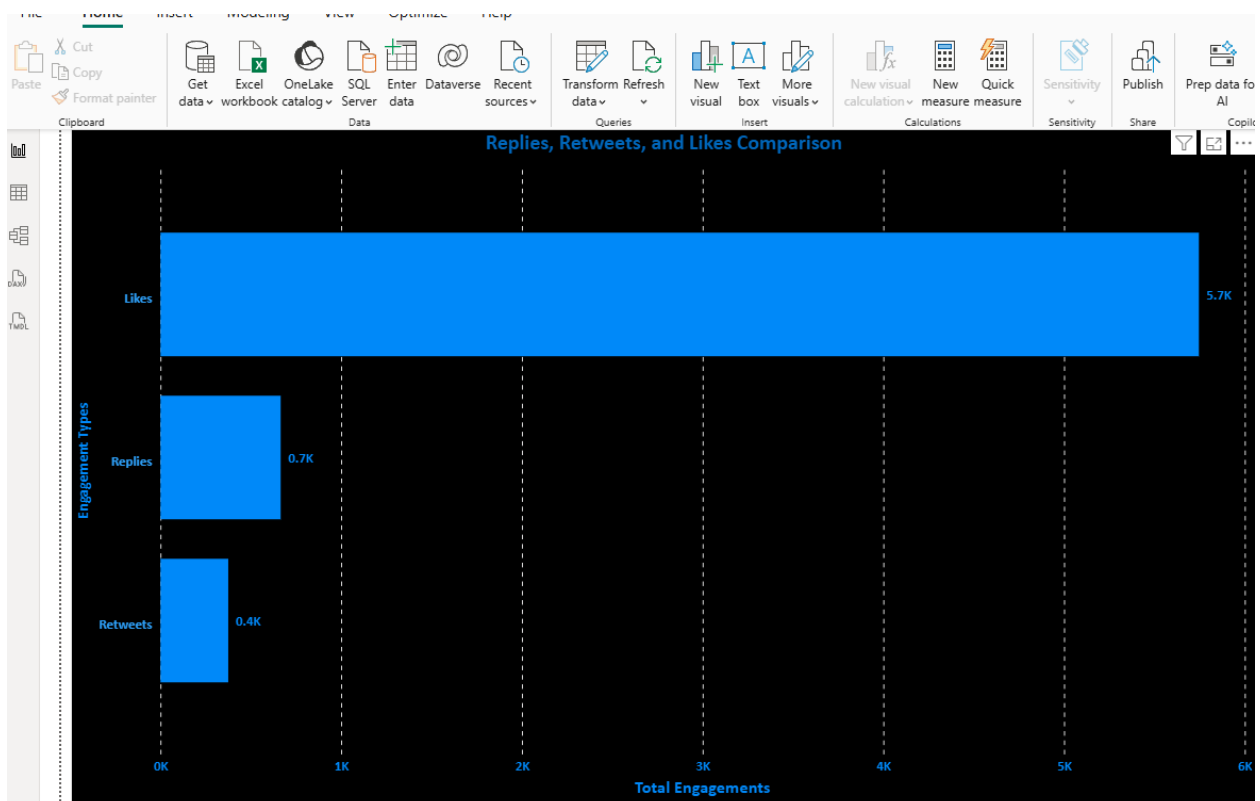
Replies, Retweets, and Likes Comparison Develop a simple visualization (bar chart) that compares the number of replies, retweets, and likes for tweets. Add a filter to include only tweets posted between June and August 2020. Use basic aggregation such as SUM for replies, retweets, and likes. This task is beginner-friendly and helps understand overall engagement metrics without complex filtering or logic.

Visualization Used

- Clustered Bar Chart

Key Insights

- Likes represented the largest share of interactions.
- Retweets contributed significantly to tweet reach.
- Replies were comparatively lower than likes and retweets.
- The comparison provides a clear understanding of user engagement patterns



Dashboard 5

Objective

To analyze the monthly trend of average engagement rate and identify periods of increased or decreased audience interaction.

Problem statement

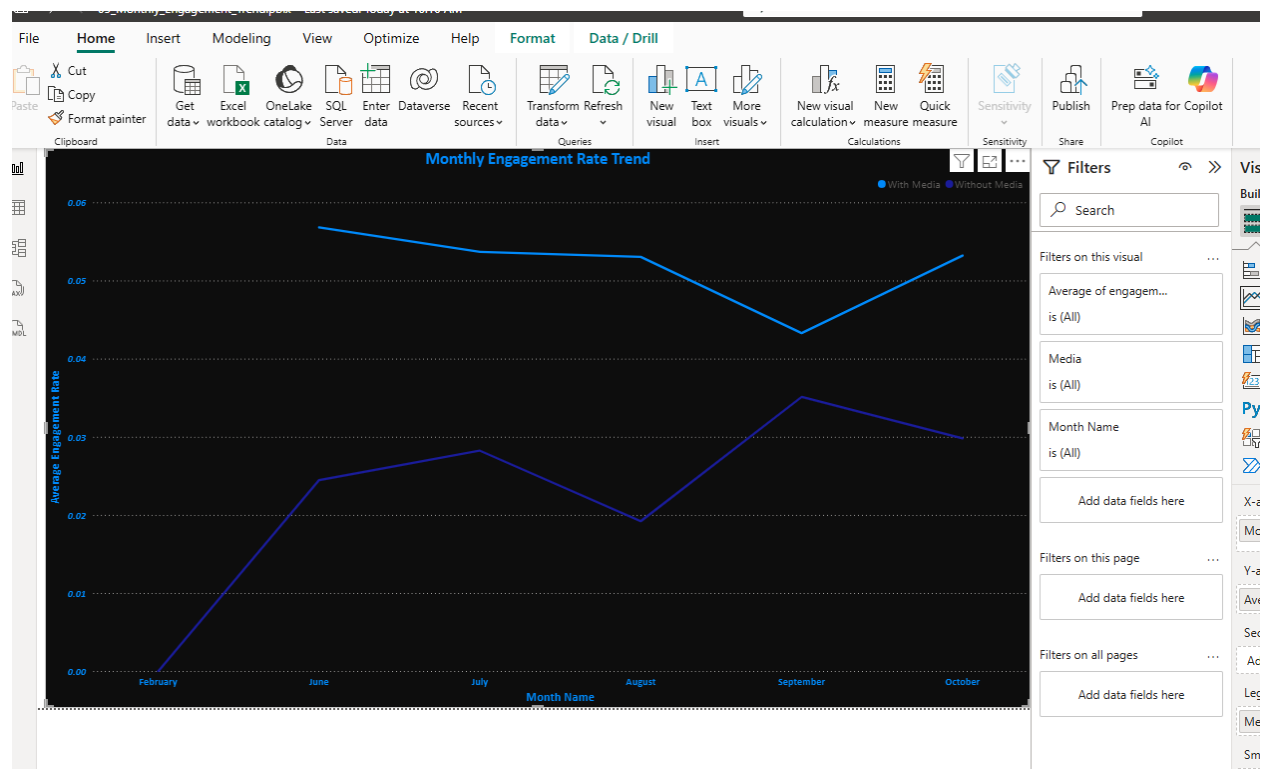
Monthly Engagement Rate Trend Create a line chart that shows the trend of average engagement rate for each month of the year. Separate the data into two lines: tweets with media content and tweets without media content. Use month as the time dimension and average engagement rate as the metric. This task is simple and helps beginners understand trends and comparisons over time.

Visualization Used

- Line Chart

Key Insights

- Average engagement varied from month to month.
- Some months experienced higher engagement than others.
- The trend analysis highlights seasonal changes in user activity.
- Monthly insights support better content planning.



Dashboard 6

Objective

To identify the top 10 tweets with the highest total engagement and analyze their overall performance.

Problem statement

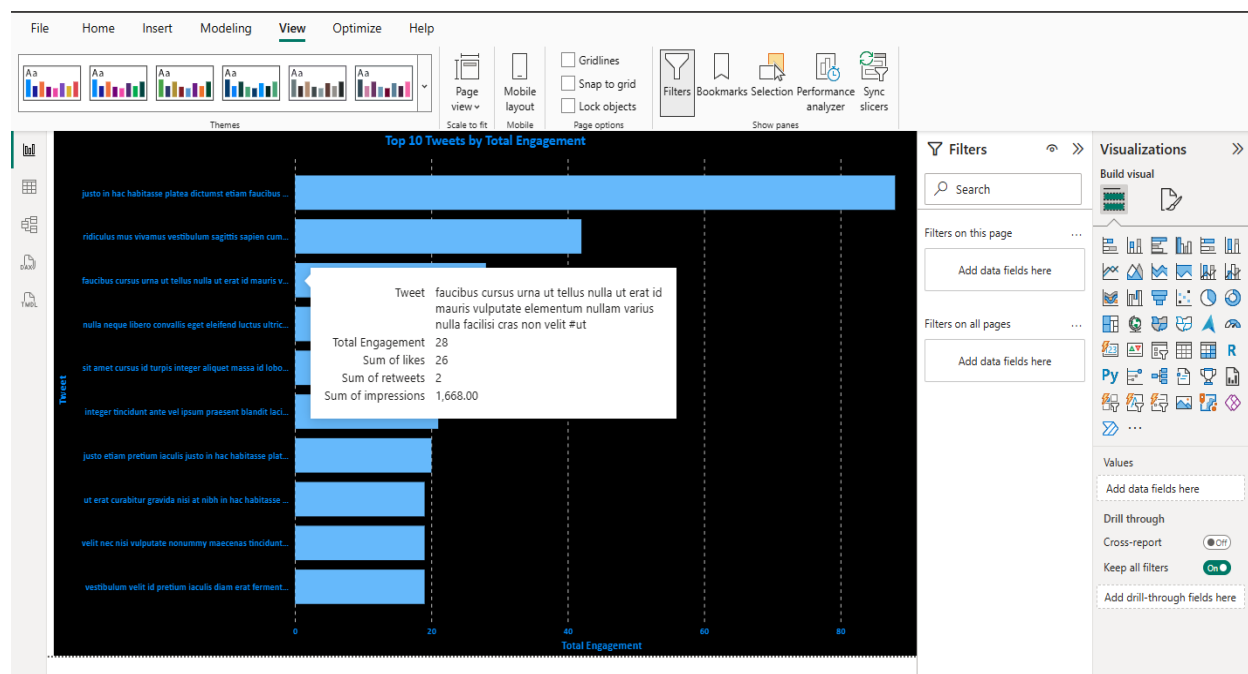
Top 10 Tweets by Engagement Build a chart to identify the top 10 tweets based on the sum of retweets and likes. Exclude tweets posted on weekends and display the user profile associated with each tweet. The chart should only be visible between 3 PM IST and 5 PM IST. Apply additional filters where tweet impressions must be even numbers, tweet dates must be odd numbers, and tweet word count must be less than 30. This task helps identify high-performing tweets and their creators.

Visualization Used

- Bar Chart

Key Insights

- The dashboard ranks tweets based on engagement.
- A small number of tweets generated the majority of interactions.
- High-performing tweets can be used as examples for future content strategies.
- Ranking helps identify the characteristics of successful tweets.



Conclusion

This project successfully demonstrates the use of **Power BI** to analyze and visualize social media engagement data through interactive dashboards. The data was prepared using **Power Query**, and **DAX measures** were used to perform the required calculations and create meaningful visualizations.

The six dashboards provide valuable insights into tweet interactions, engagement rates, media performance, monthly engagement trends, and top-performing tweets. These visualizations make it easier to identify user behavior patterns, compare engagement metrics, and monitor content performance effectively.

Overall, this project enhanced my practical knowledge of **data cleaning, data modeling, DAX, and dashboard design** in Power BI. It also strengthened my ability to transform raw data into actionable business insights, demonstrating how business intelligence tools can support informed decision-making.